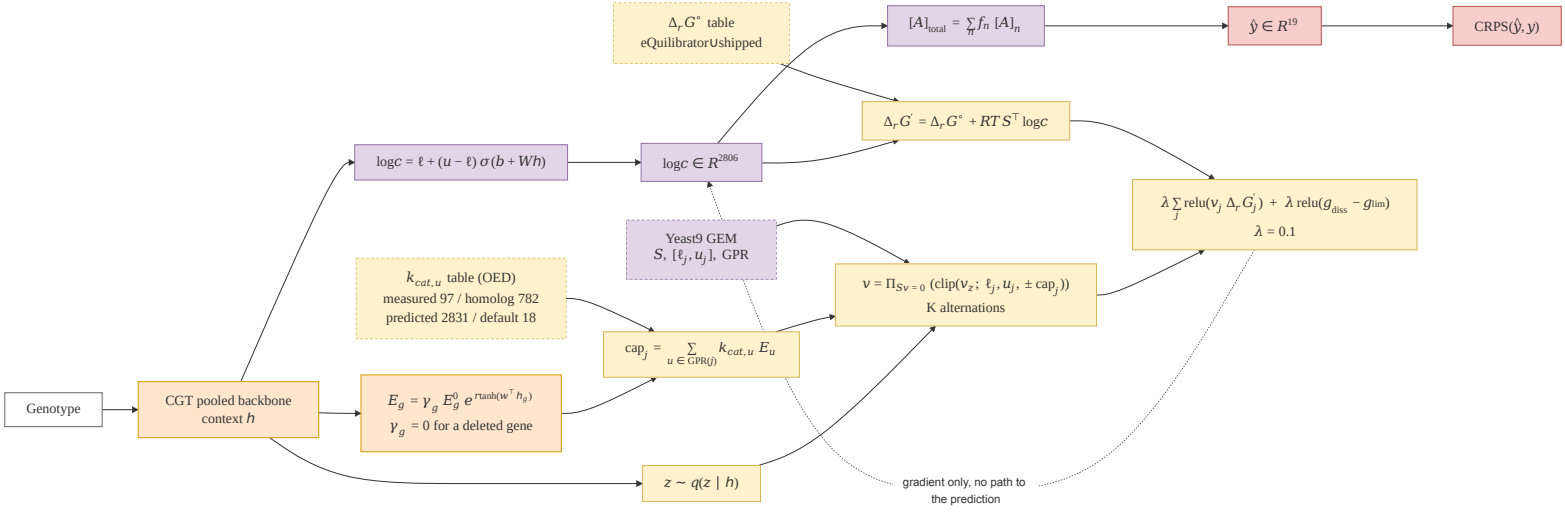

Genotype
CGT pooled backbone context h
$\log c = \ell + (u - \ell)\,\sigma(b + Wh)$
$\log c \in R^{2806}$
$\Delta_r G^\circ$ table
eQuilibratorUshipped
$[A]_{\text{total}} = \sum_n f_n\,[A]_n$
$\hat{y} \in R^{19}$
CRPS($\hat{y}, y$)
$\Delta_r G' = \Delta_r G^\circ + RT\,S^\top \log c$
$\lambda \sum_j \text{relu}(v_j\,\Delta_r G'_j) + \lambda\,\text{relu}(g_{\text{diss}} - g_{\text{lim}})$
$\lambda = 0.1$
Yeast9 GEM
S, $[\ell_j, u_j]$, GPR
$k_{cat,u}$ table (OED)
measured 97 / homolog 782
predicted 2831 / default 18
$\text{cap}_j = \sum_{u \in \text{GPR}(j)} k_{cat,u}\,E_u$
$v = \Pi_{Sv=0}\,(\text{clip}(v_z;\ \ell_j, u_j, \pm \text{cap}_j))$
K alternations
$E_g = \gamma_g\,E_g^0\,e^{r \tanh(w^\top h_g)}$
$\gamma_g = 0$ for a deleted gene
$z \sim q(z \mid h)$
gradient only, no path to the prediction